

Anticipating Outbreaks: Integrating Multi-Source Signals for Early Detection and Decision Support

CSE 6242: Team 145 Final Report

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1. Introduction

This project complements traditional infectious disease monitoring and reporting systems by integrating data across countries to allow for a unified, cross-border approach to outbreak detection and response. By combining formal surveillance with informal signals, such as news, social media, search activity, and logistics trends, we aim to identify emerging health events earlier and more consistently than official channels and encourage active intervention. Our goal is to develop a practical, scalable early-warning and decision-support system for Southeast Asia that turns routinely available data into real-time, actionable insights and supports more coordinated global action.

2. Problem Definition

Disease detection and reporting across Southeast Asia remain hindered by fragmented systems, inconsistent data pathways, and substantial delays in information flow. Most countries depend on hierarchical reporting structures in which local case information moves upward through district and national authorities, with only select officials authorized to escalate potential threats (Talwar et al., 2026). Although some nations use open-source media monitoring or automated Bayesian tools for preliminary signal detection, these systems still require manual verification (Magid et al., 2018). Reporting delays, incompatible data formats, and uneven surveillance capacity further slow response efforts (Amit et al., 2021). These challenges are compounded by gaps in laboratory-based pathogen detection: shortages in funding, trained staff, and equipment create patchwork surveillance coverage and leave rural or low-resource settings especially vulnerable (Getchell et al., 2024).

At the same time, a wealth of digital data—search trends, global news coverage, mobility signals, supply-chain indicators, and online reports—remains underused because it is scattered across platforms, difficult to preprocess, or too technical for non-specialists to interpret. As a result, health ministries, hospitals, humanitarian responders, and logistics teams often lack timely, consolidated insights for making informed decisions. The groups most harmed by these delays include rural communities, migrant workers, and under-resourced clinics, all of whom face the highest risks when detection lags behind disease spread.

3. Literature Survey

Across the literature we surveyed, the main idea in predicting and preventing pandemics is to move from *reactive* outbreak response to *anticipatory* preparedness by combining diverse early signals with explicit decision thresholds (“triggers”) that activate predefined actions (Alcayna et al., 2025). This is useful for our project because it supports an end-to-end design: (1) broaden detection beyond routine case reporting using informal digital and event-based systems (Magid et al., 2018; Hutchinson et al., 2024a, 2024b), search behavior that tracks dengue dynamics (Husnayain et al., 2019; Ho et al., 2018), crowdsourced inputs and tailored communication (Lwin et al., 2014), and pharmacy sales as syndromic proxies (Wulandari et al., 2022); (2) improve forecast localization using mobility data to guide targeted readiness and pre-positioning (Kiang et al., 2021); and (3) connect surveillance outputs to operational constraints such as data sharing, reporting incentives, laboratory capability, and supply vulnerability (Amit et al., 2021; Talwar et al., 2026; Getchell et al., 2024; Das et al., 2022), grounded against routine reporting products (WHO, 2025; Communicable Diseases Agency, 2026). The main shortcomings to manage are biased or non-representative signals (media/search/crowdsourcing reflect

attention and access, not incidence) (Ho et al., 2018; Magid et al., 2018; Lwin et al., 2014), privacy/access barriers and model drift for mobility-linked data (Kiang et al., 2021), confounding and stockout effects in pharmacy sales (Wulandari et al., 2022), uneven genomic validation and workforce and procurement dependencies (Getchell et al., 2024; Stona et al., 2026), cross-border interoperability and political/bureaucratic disincentives to report (Amit et al., 2021; Talwar et al., 2026), and broad “big data” reviews with limited empirical validation (Olaboye et al., 2024).

4. Proposed Method

Current outbreak detection relies on official, indicator-based surveillance: clinicians identify cases, labs confirm diagnoses, and results move through reporting chains before alerts are issued. While essential for verification, this process is often slow, especially in lower-resource settings, due to dependencies on care-seeking behavior, testing capacity, and administrative workflows. This creates a critical lag between early community transmission and actionable signals.

While leveraging informal data such as social media as early warning detection sources is not novel in concept, attempts to integrate this method into traditional monitoring systems have had mixed results as informal data sources are inherently noisy, disparate, and difficult to access. Our approach addresses these issues by focusing on more reliable, accessible signals such as national news sources and targeted search trends. We further validate these signals with logistics/mobility patterns and health data. By requiring cross-signal agreement and applying adaptive weighting by location and season, the system reduces false alarms while surfacing risks earlier than traditional reporting.

The data used for the model development and evaluation emanated from different sources: official reports setting the ground truth (WHO), demographic data (UN), news outlets, social media, Google searches, and International Monetary Fund. A summary of the data sources is provided below.

Table 1: Overview of Data Sources Used in the Early Warning System

Type	Source	Method	Traits	Key Variables
Outbreak reports, demographics	WHO, UN	Manual data file download	114 KB, 3168 records, 38 features	ISO week; country; disease; case and death counts; linkable identifiers for matching to signals; mobility_index; ihr_score; ihr_beds_per_10k; physicians_per_10k; pop_density; urban_pct; temp; humidity
News, social media, and internet search behavior	Online news outlets, Google Trends	API – GDELT, web scraping	8.7 MB, 2579 records, 86 features	Article timestamp and ISO week; headline/body text; location mentions; disease/symptom keywords; count of relevant articles; and sentiment
Logistics & mobility	World Bank and IMF	Manual data file	100 MB, 262k	Goods and services volume trades; deviations from baseline mobility;

		download, API – IMF	records, 57 features	connectivity features (in-/out- flow).
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We built our modeling pipeline around a simple idea. Official reports from WHO Disease Outbreak News (WHO DON) are not perfect, but they provide a consistent reference point for when an outbreak becomes visible in formal surveillance. We therefore used WHO DON as our outcome signal, and we trained classification models (from data between January 2018 and December 2022) to estimate outbreak risk individually using health, media, and logistics data sources and then collectively from a combination of the three with a top-k features approach. The test dataset comprised of data between January 2023 and December 2025.

Our target was a four-level monthly risk label for each Southeast Asian country–month–disease combination. Instead of using WHO DON report counts alone, we adjusted the risk labels using available DON reports and observed outbreak severity. For each country, month, and disease, we calculated a monthly severity score based on reported cases and death. Deaths were weighted more heavily because they better reflect outbreak severity and are typically lower in count than reported cases. For each country–disease pair, monthly severity scores were ranked and converted into four ordered labels: low, moderate, high, and critical. This transformed irregular outbreak reporting into a consistent monthly training signal from January 2018 through December 2025.

$$Severity\ Score = \log(1 + reported\ cases) + 2\log(1 + deaths)$$

$$Risk\ Label = \begin{cases} Low & 1^{st} \leq Percentile_{Sev\ Score} \leq 25^{th} \\ Moderate & 25^{th} < Percentile_{Sev\ Score} \leq 50^{th} \\ High & 50^{th} < Percentile_{Sev\ Score} \leq 75^{th} \\ Critical & 75^{th} < Percentile_{Sev\ Score} \leq 99^{th} \end{cases} \text{ if}$$

We then aligned all data sources to the same country–month–disease structure and transformed them into model-ready features. Four classification models were developed: a Health/Epi model using outbreak and demographic baseline features, a Media/Web model using news, social media, and search signals, a Logistics model using supply chain, mobility, and logistics indicators, and a Hybrid model using the top 10 selected features from the three individual models. Due to data availability, train and test evaluation focused on 10 ASEAN countries, 3 diseases—Avian Influenza, Dengue, and COVID-19—and 96 months from January 2018 to December 2025.

We then use these scores in an interactive platform that shows the risk level for a selected location and month via an alert system. It is delivered through an interface that enables users to rapidly identify emerging areas of concern and understand current and previous risk and vulnerabilities in each location. The platform is designed for public health teams, hospital planners, humanitarian responders and logistics partners who require timely insight to inform decision-making, resource distribution and operational readiness.

The interface consists of two map visualizations designed to be viewed sequentially. The first map visualizes early signals derived from curated news sources and search trend data. Content is filtered

using disease- and infection-related keywords, and referenced cities are plotted geographically. Each referenced city appears as a red circle with a radius proportional to the volume of relevant articles.

The second map presents model-generated risk classifications at the country level. Users can select between models of risk (Hybrid, Health, Logistics, Media, Actual), and this view is implemented using an interactive visualization built in Observable. From an intuition perspective, Observable makes it easier to show complex data in a simple and interactive way compared to static charts. Instead of looking at multiple separate graphs, users can switch between models and time periods in one place. This helps users quickly focus on what they need and understand changes over time.

For the implementation, the map is built using D3.js inside Observable. A world map (GeoJSON) is loaded, and each country is colored based on its risk level from the dataset. The data is organized by country and month, with different columns for each model. A dropdown menu lets users choose which model to view, and a time slider (year and month) updates the map dynamically. When the user changes these options, the map automatically updates using Observable's reactive system. Tooltips are also added so users can see details like country name and risk level when hovering. Missing data is shown in gray to avoid confusion.

Both visualizations contain a dynamic time filter (year and month) to allow users to explore how signals and risk levels evolve across space and time. Together, these two views support a continuous monitoring workflow, pairing early signal detection with synthesized risk assessment.

5. Evaluation

Study setting: The testbed is an ASEAN country-level outbreak early warning setting, with observations aligned by month and risk outputs expressed in four classes: *Low*, *Moderate*, *High*, and *Critical*. The countries represented in the reported results are Thailand, Lao PDR, Cambodia, Myanmar, Singapore, Vietnam, Brunei, Malaysia, Indonesia, and the Philippines.

Prediction task: The task is multi-class classification of outbreak risk level relative to each country's baseline. The purpose is to identify elevated risk earlier than routine reporting alone.

Model configurations under test: Four configurations for classification of risk were evaluated: 1) EPI (Health): Logistic Classifier; 2) WEB (Media): Extra Trees Classifier; 3) LOG (Logistics): Random Forest and 4) HYB (Hybrid): Random Forest.

Evaluation measures: Two measures were used: 1) AVG_ADRL (Average Absolute Difference in Risk Label): lower values indicate better agreement with observed risk labels; 2) Match Rate: higher values indicate better exact class agreement.

Experimental questions: The experiments were designed to answer: 1) Which model configuration performs best overall? 2) Does a hybrid configuration outperform single-source configurations? 3) How does performance vary by disease type? 4) How does performance vary by country? 5) What is the effect of sparse or missing data streams on model performance?

Experiment A: Overall model comparison

Model	Best Algorithm	Sample Size	AVG_ADRL	Match Rate
EPI	Logistic Classifier	562	0.71	0.55
WEB	Extra Trees Classifier	479	0.68	0.57
LOG	Random Forest	479	0.65	0.58
HYB	Random Forest	479	0.83	0.48

Observations:

1. LOG is the strongest overall configuration (lowest AVG_ADRL, highest Match Rate).
2. WEB ranks second and is close to LOG.
3. EPI is third and remains workable where richer streams are unavailable.
4. HYB performs worst on both measures and does not improve results in this testbed.

Experiment B: Disease-level performance

Disease	Sample Size	Overall AVG_ADRL	EPI	WEB	LOG	HYB
Dengue	1031	0.60	0.65	0.53	0.55	0.65
Avian Influenza	228	0.65	0.63	0.60	0.54	0.85
COVID-19	740	0.90	0.81	0.92	0.81	1.08

Observations:

1. COVID-19 is the most difficult disease in this dataset (highest error values).
2. Dengue is best captured by WEB (0.53), with LOG close behind (0.55).
3. Avian Influenza is best captured by LOG (0.54).
4. For COVID-19, EPI and LOG tie as the best-performing configurations (0.81).
5. HYB is consistently weak across all disease groups.

Experiment C: Country-level performance

Country	Sample Size	Overall AVG ADRL	EPI	WEB	LOG	HYB
Thailand	208	0.55	0.58	0.56	0.50	0.56
Lao PDR	39	0.56	0.56	N/A	N/A	N/A
Cambodia	228	0.62	0.60	0.70	0.51	0.68
Myanmar	304	0.71	0.79	0.66	0.59	0.79
Singapore	212	0.71	0.79	0.49	0.58	0.96
Vietnam	284	0.72	0.62	0.72	0.69	0.85
Brunei	44	0.73	0.73	N/A	N/A	N/A
Malaysia	168	0.74	0.52	0.64	0.79	1.00
Indonesia	232	0.82	0.88	0.79	0.76	0.84
Philippines	280	0.87	0.87	0.86	0.77	0.97

Observations:

1. LOG is best in most countries (Thailand, Cambodia, Myanmar, Indonesia, Philippines).
2. EPI is best in Lao PDR, Brunei, Vietnam, and Malaysia, indicating value under sparse data conditions.
3. WEB is best in Singapore.
4. HYB is never best and is substantially weaker in several countries (for example Malaysia, Singapore, Philippines).
5. Lowest overall country error is Thailand (0.55); highest is the Philippines (0.87), followed by Indonesia (0.82).

Experiment D: Data availability and robustness

Observations:

1. Missing values for WEB/LOG/HYB in Lao PDR and Brunei indicate incomplete stream coverage.

2. Country sample sizes vary widely (for example $n=39$ in Lao PDR vs $n=304$ in Myanmar), which affects stability of country-level estimates.
3. Where data are sparse, EPI remains usable and provides a practical fallback.
4. Performance patterns suggest that mobility/logistics indicators are often strong early signals, but benefits depend on data completeness and quality.

Across the reported experiments, the LOG configuration (Random Forest on logistics/mobility-related features) provides the best overall performance in this testbed. The EPI configuration is the most reliable fallback where non-health data streams are missing or incomplete. The HYB configuration, as implemented in this study, does not improve predictive performance and should not be selected as the default model in its current form.

6. Conclusions and Discussion

This project developed an ASEAN-focused early warning system that combines official outbreak records with faster proxy signals from health, media/web, logistics, and mobility data to estimate monthly pandemic risk when official reporting is limited.

Traditional WHO DON and regional health reports are valuable, but they can be sporadic, retrospective, and unavailable for many country–disease–month combinations. By using routine multi-source signals, the system expands situational awareness beyond traditional surveillance and supports more timely, interpretable outbreak monitoring.

Key observations:

- The logistics-based model performed best overall among the tested approaches.
- The health/epidemiological model remained a useful fallback when informal data was unavailable.
- The hybrid model did not improve performance in its current form, suggesting that combining data sources requires more careful feature selection.
- Interactive dashboard maps made risk patterns easier to compare across countries, diseases, months, and models than the static, standalone tabular reports released by WHO.

Limitations and future work:

- Data availability was uneven across countries, diseases, and informal data streams.
- Future work should improve API/data-sharing access, missing-data handling, label definitions, prospective validation, and hybrid model refinement.

Overall, the results show that non-health signals—such as news coverage, search behavior, mobility, and logistics activity—can provide more timely outbreak insight, especially for vulnerable populations in areas with less frequent official reporting. The findings also suggest that different data sources may be more informative for specific countries, diseases, or time periods, highlighting the value of a flexible multi-source approach. More broadly, this research demonstrates how algorithmic modeling and interactive visualization can transform complex, fragmented data into clearer, more consistent, and more actionable pandemic risk insights.

Team Effort Statement

All team members contributed a similar amount of effort across data preparation, model development, evaluation, and dashboard/report preparation.

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